

EE 555: Intelligent Control System

Course Code	EE 555
Course Name	Intelligent Control System
L-T-P-C	3-0-0-3
Prerequisites	For B.Tech. – Control System or equivalent course; For all – Linear Algebra
Intended for	B.Tech. EE (3rd and 4th year), M.Tech., M.Tech. (R), PhD
Mutual Exclusion	None
Approval	54th BoA (May 2024)

Course Contents

- **Nonlinear Control - Primer:** Norms, Sign-definiteness, State-space model, Lyapunov stability theory, Discrete-time systems, Nonlinear control strategies. (4 Hours)
- **Neural Network:** Feed-forward networks, Multi-layered neural networks, Radial-Basis function networks, Feedback networks, System identification using neural network (6 Hours)
- **Fuzzy Logic:** Classical sets, Fuzzy sets, Approximate reasoning, Fuzzy logic control, System identification using Fuzzy models (6 Hours)
- **Indirect Adaptive Control using Neural Networks:** Continuous-time affine systems, Discrete-time affine systems, Discrete-time non-affine systems (6 Hours)
- **Direct Adaptive Control using Neural Networks:** Direct Adaptive control, Single-input single-output affine systems, Multi-input multi-output systems, Backstepping control (6 Hours)
- **Reinforcement Learning (Approximate Dynamic Programming):** Linear quadratic regulator, HJB formulation, HJB for affine systems, Heuristic and Dual heuristic dynamic programming, Single network adaptive critic, Continuous-time adaptive critic (8 Hours)
- **Intelligent Control of Benchmark Systems:** Pendulum on cart, Mobile robot, Robot manipulator, HVAC system (6 Hours)

Laboratory Modules

- None

Text Books

1. Ali Zilouchian and Mo Jamshidi, **Intelligent Control Systems using Soft Computing Methodologies**, CRC Press.
2. Dusko Katic and Miomir Vukobratovic, **Intelligent Control of Robotic Systems**, Springer.
3. Laxmidhar Behera and Indrani Kar, **Intelligent Systems and Control: Principles and Applications**, Oxford University Press.

References

1. Thrisantha Nanayakkara, Ferat Sahin, and Mo Jamshidi, **Intelligent Control Systems with an Introduction to System of Systems Engineering**, CRC Press.